



Course name: INTERMEDIATE MACROECONOMICS
Course code: EC2211
Type of exam: Main
Examiner: Martin Flodén
Number of credits: 7.5
Date of exam: Thursday 30 October 2025
Examination time: 5 hours (08:00-13:00)
Aids: No aids are allowed

Write your anonymization code on each answer sheet (the code you received from Ladok when you registered for the exam).

Start each new question on a new answer sheet.

Explain notions/concepts and symbols. If you think that a question is vaguely formulated, specify the conditions used for solving it. Only legible exams will be marked.

The exam consists of 5 questions, worth 100 points in total. The maximum number of points on each sub-question are stated within parenthesis. For the grade E 45 points are required, for D 50 points, C 60 points, B 75 points and A 90 points.

All questions should be answered in English.

Your results will be made available on your Ladok account (www.student.ladok.se) within 15 working days from the date of the examination.

Good luck!



Question 1: Short questions (25 points in total)

Please write no more than one page (maximum) when answering each of these questions.

- a. Consider an economy where the production function is $Y_t = A_t K_t^\alpha L_t^{1-\alpha}$ where A , K , and L denote TFP, capital and labor, respectively. Suppose that $\alpha = 1/3$ and that the growth rates of output and the capital stock are $g_Y = 1.5\%$ and $g_K = 3\%$ while the population is constant. Calculate the TFP growth rate. (5 points)
- b. Consider the effects of a fiscal stimulus. It is often argued that the fiscal stimulus will have larger effects on output and inflation when the monetary policy rate is at its lower bound. Explain the logic behind this argument. (5 points)
- c. Explain what is meant by Ricardian equivalence. Do you think that this theory is likely to hold up in reality? Why/why not? (5 points)
- d. Consider an individual who lives for a single period, chooses how much to work and who has no other income than the wage income. She derives utility from consumption and disutility from working according to the following utility function:

$$U(C, H) = \ln(C) - H^\gamma$$

Where C is consumption, H is hours worked and $\gamma > 0$ is a parameter. Let w denote the wage rate. Formulate the individual's budget constraint and solve for the individual's optimal choices of consumption and hours worked. (5 points)

- e. In the years after World War II, when much of Europe lay in ruins, the United States ran a trade surplus. Use your knowledge about how the trade balance relates to international capital flows to provide a hypothesis that could explain what was going on. (5 points)

Question 2: The Solow model (25 points)

The population growth rate has fallen in many countries over the last decade. Your task is to use the Solow model to analyze the consequences of a fall in the population growth rate. Consider therefore a version of the Solow model where the population, L_t , grows at rate n . A fraction s of income is invested in capital every period, and capital depreciates at rate δ . The production technology is Cobb-Douglas and given by:

$$Y_t = AK_t^\alpha L_t^{1-\alpha}$$

where TFP, A , is constant and $0 < \alpha < 1$.

- Derive an expression for the accumulation of capital per person in this economy, i.e., Δk_{t+1} , where $k_t \equiv K_t/L_t$. (4 points)
- What is the steady state condition in this economy? Illustrate the equilibrium in a diagram. (4 points)
- Suppose that the economy starts out in steady state. Analyze the effects of an immediate and permanent fall in the population growth rate. Use the Solow model to explain what happens to the economy over time and in the long run. In particular, comment on what happens to the level and growth rate of per capita GDP. (12 points)
- Consider now instead the Romer model. Briefly explain how and why the conclusions from part (c) would change according to this model (do not use equations or graphs). (5 points)

Question 3: Aggregate supply and demand (20 points)

Consider an economy that begins in period $t = 0$ with output at potential, $\tilde{Y}_0 = 0$, and the inflation rate at the target, $\pi_0 = \pi_0^*$. In the beginning of period $t = 1$, a new chair of the central bank announces a new lower inflation target $\pi_t^* = \bar{\pi}$ for all periods $t \geq 1$ (where $\bar{\pi} < \pi_0^*$).

- Summarize the main assumptions behind the AS and AD curves. (4 points)
- Use the AS-AD model to show how the announcement of the new inflation target affects the economy in periods $t = 1$ and $t = 2$. In answering this question, assume that inflation expectations are adaptive and given by $\pi_t^e = \pi_{t-1}$. (8 points)
- Suppose now that inflation expectations are partially influenced by the central bank's inflation target and partially adaptive. More specifically, suppose that $\pi_t^e = \alpha\pi_t^* + (1 - \alpha)\pi_{t-1}$ where $0 < \alpha < 1$. Would the transition to the new steady state then be different? Again, use the AS-AD model to show how the economy responds in periods $t = 1$ and $t = 2$. (8 points)

Question 4: The Great Inflation (20 points)

This is an essay question where you are supposed to refer to theories and material covered in the course but you need not use equations or graphs in your answer. Write no more than 3 pages (maximum). Only legible answers will be considered.

Inflation increased rapidly in the United States and many other countries during the 1970s. The oil price shocks and other adverse supply shocks are often cited as explanations for this surge in inflation. In the course, however, we have discussed several additional factors that may also have contributed to the rise in inflation during the 1970s.

- List three such factors and briefly explain how each may have contributed to higher inflation.
- Inflation also increased rapidly in the United States and Europe in 2021 and 2022 but has since returned to more normal levels. Based on material covered in the course, discuss briefly why the recent inflation surge may have been less persistent than in the 1970s.



Question 5: The time consistency problem (10 points)

This is a credit question. Students with credit from the exercises will be awarded the better of that credit and the result on this question. I suggest that you only answer this question if you lack sufficient credit from the exercises.

Consider an economy with only one time period. In the beginning of the period, households form inflation expectations π^e . After inflation expectations have been formed, the policymaker chooses the inflation level π . Finally, the level of output is realized from the Phillips curve:

$$y = \bar{y} + \kappa(\pi - \pi^e)$$

where $\bar{y}$ is potential output.

Suppose that the policymaker chooses π to hold inflation as close to zero as possible and output as high as possible. More precisely, she chooses π to minimize this loss function:

$$L(\pi, y) = \pi^2 - \lambda y$$

where λ is a parameter that indicates how much the policymaker cares about economic activity.

- Find an expression for π as a function of π^e , $\bar{y}$, and the parameter values. (4 points)
- When forming expectations, households understand the policymaker's problem and anticipate her behavior. Therefore, $\pi^e = \pi$. What will then be the equilibrium outcome for π , π^e , and y ? (4 points)
- Very briefly, explain why the parameter λ affects the equilibrium outcome. (2 points)